

Snehal Gore

Los Angeles, CA | ssgore@usc.edu | +1 310-462-7354 | [linkedin.com/in/snehal-gore](https://www.linkedin.com/in/snehal-gore) | github.com/snehalgore1

SUMMARY

Software Engineer with 2.5+ years building and shipping production systems, graduating with an M.S. in Computer Science from USC in December 2026. Experience in backend services, distributed systems, ML infrastructure, and performance-critical C++/CUDA software. Shipped software across 1M+ devices at Whirlpool and built production backend and AI systems at an early-stage InsurTech startup.

EXPERIENCE

SnapRefund (Early-Stage InsurTech Startup)

Los Angeles, CA

Software Engineer Intern

May 2026 - Aug 2026

- Architected and deployed AiSnap, a production AI microservice using Gemini function calling over live PostgreSQL data spanning **112 ORM entities**; exposed database-grounded tools through a NestJS API layer.
- Built a validation and grounding pipeline that detected responses lacking query provenance and triggered tool-grounded regeneration, improving reliability for financial queries.
- Built and shipped backend features end to end, including **database migrations, administrative workflows, and payment-processing logic** integrated into the production billing pipeline.
- Debugged and shipped features across backend and frontend services through **test, staging, and production** environments in a fast-moving startup environment.

Whirlpool Corporation

Pune, India

Software Engineer, Machine Learning

Jul 2022 - Dec 2024

- Built and deployed production inference frameworks across **1M+ smart appliances**, eliminating **\$250K+/year** in cloud inference costs while delivering **3× lower latency**.
- Owned production systems from development through deployment, including model optimization, automated pipelines, **OTA rollouts, field evaluation, and production monitoring**; mentored interns on development and deployment workflows.
- Built large-scale time-series data and training pipelines on **AWS Redshift**, reducing dryer Estimated Time Remaining error **7× from 35 to 5 minutes**.
- Led productionization of a computer-vision system, shipping a **FastAPI inference service** and automating deployment and lifecycle management with AWS SageMaker and MLflow.

PROJECTS

Distributed Object Store | github.com/snehalgore1/distributed-object-storage

C++20, gRPC, Raft, Protobuf, SQLite, CMake, Docker, Kubernetes

- Built a fault-tolerant distributed object store in **C++20** with consistent hashing, **3-way replication and quorum writes**, checksum-verified reads, failure detection, and anti-entropy repair across storage nodes.
- Implemented a **Raft consensus engine from scratch** for the metadata control plane, including pre-vote elections, persistent replicated logs, majority commit, follower catch-up, and leader failover over gRPC.
- Engineered concurrent request handling with **bounded thread pools, sharded read/write locks, WAL-backed crash recovery, and atomic fsync/rename writes**; validated with **131 unit/integration tests and ThreadSanitizer CI**, reaching **~3.8K requests/sec** in distributed benchmarks.

MiniTensorRT: C++ Transformer Inference Runtime | github.com/snehalgore1/mini-tensorrt

C++17, CUDA, ARM NEON SIMD, CMake, ONNX

- Built a from-scratch **C++17 inference runtime** for GPT-2 (124M) with HuggingFace output parity; optimized GEMM from **1.7 to 310 GFLOP/s (~180×)** using ARM NEON SIMD and multithreading, reducing transformer-block latency **56×** and memory usage **12.5×** through arena memory planning and operator fusion.
- Extended the runtime to **CUDA**, running GPT-2 in **~22 ms** on a Tesla T4; implemented FP16 Tensor Core and tiled SGEMM kernels, KV-cache decoding (**1.87× speedup**), FlashAttention-style online softmax, INT8 quantization, and ONNX model support.

Agentic RAG & Information Retrieval Platform | github.com/snehalgore1/agentic-rag

Python, LangGraph, FastAPI, OpenSearch, PostgreSQL, Redis, Airflow, Docker

- Architected an end-to-end RAG platform indexing **10K+ documents across 7 containerized services**, serving grounded responses at **<2s p95 latency**; combined BM25 and dense retrieval using Reciprocal Rank Fusion, improving Recall@10 by **35%**.
- Built an adaptive retrieval workflow with relevance grading, query rewriting, retrieval routing, and out-of-domain guardrails using **FastAPI, OpenSearch, PostgreSQL, Redis, Airflow, and Docker**.

EDUCATION

University of Southern California

Los Angeles, CA

M.S. Computer Science, AI Specialization

Jan 2025 - Dec 2026 (Expected)

Coursework: Analysis of Algorithms, Web Technologies, Information Retrieval & Web Search Engines, Hardware Foundations of Machine Learning, Machine Learning, Applied NLP, Deep Learning, Foundations of AI

Savitribai Phule Pune University

Pune, India

B.E. Computer Engineering

May 2018 - May 2022

TECHNICAL SKILLS

Languages: C++, Python, Java, TypeScript, JavaScript, SQL

Systems: Distributed Systems, Multithreading, Concurrency, gRPC, Protobuf, Raft, CUDA, ARM NEON SIMD, CMake

Backend & Data: FastAPI, NestJS, Node.js, PostgreSQL, Redis, OpenSearch, MySQL, REST APIs

Cloud & Infrastructure: AWS, GCP, Docker, Kubernetes, GitHub Actions, Airflow, Pulumi

ML Systems: PyTorch, TensorFlow, ONNX, MLflow, SageMaker, model optimization, quantization